MDLM few-step generation: `marginal` vs `edlm` at k tokens per step

QED / fewstep

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Read this before quoting any hits number in this report.

Every hits@0.6 figure below is counted on `novel`, defined as set(valid)$\setminus$QM9, and that metric is **confounded by molecule size in a way that scales with** k. QM9 contains only molecules of ≤ 9 heavy atoms; RDKit validity does not enforce that, and QED rewards size (it peaks near 300 Da, far above anything QM9 can reach). Two biases therefore compound and both grow with k: molecules outside the size range are novel *by construction*, and they score higher QED. Measured on $N = 1000$, $\lambda = 200$, `edlm`: mean heavy-atom count runs $9.0 \to 13.6$ and the fraction above 9 heavy atoms runs $38.9\,\% \to 95.7\,\%$ as $k : 1 \to 32$.

Recounted on metrics that are not size-confounded (5-seed means, same runs):

k (at $N = 1000$, $\lambda = 200$)	1	4	8	32
as reported below (novel)	69.6	111.2	**137.4**	98.2
unique valid, no QM9 subtraction	**166.0**	123.8	139.6	98.2
restricted to ≤ 9 heavy atoms	**115.0**	30.2	11.6	0.2

Invalidated: every claim that few-step raises hits, the reward-call Pareto frontier, and the reading of novel/valid $\to 1$ as "duplication disappears" (it is the size drift). **Still valid:** the `marginal` vs `edlm` contrast (both modes drift identically, to one decimal place in heavy-atom count, so it cancels in Δ), the two null controls, the validity collapse being a property of few-step itself, and the `allbad` analysis.

Synthetic accessibility is *flat* across k (SA 4.74 / 4.74 / 4.75 at $k = 1/8/32$ against 3.61 for QM9), so the larger molecules are not unmakeable junk — this is a confounded comparison, not reward hacking.

Recount data: `results/qed/fewstep/_recount.csv` (1240 runs), produced by `scripts/ours/recount_size_controlled.py`. The choice of replacement metric is an open decision; this report has not been rewritten under it.

1 What this sweep is

At $k = 1$ unmasked position per step the two mixture samplers are the *same Markov chain*: `marginal` returns $\mathbb{E}_n[q(\cdot \mid x_t, x_0^{(n)})]$ while `edlm` draws $n^* \sim \mathrm{Cat}(w)$ and samples that component, and with one position updating the joint equals that position's marginal. The whole difference therefore lives in steps with $k \geq 2$, where `marginal` draws each position independently from the reward-weighted candidate histogram while `edlm` copies k tokens from a single candidate. This sweep puts k on an axis.

No code change was needed. `position_selection=random_k` unmasks round$(n_{\text{masked}} \cdot p_{\text{unmask}})$ positions and for the log-linear schedule $p_{\text{unmask}} = \mathrm{d}t/t$ exactly, so $k = L/T$ deterministically at every step; verified numerically for $T \in \{32, 16, 8, 4\}$ with no rounding drift and no leftover mask. Uniformly random positions are not a heuristic here: every masked position carries the same "stay masked" probability, so conditional on the count the position set is uniform over subsets — `random_k` *is* the true kernel conditioned on the count.

Arm A settings throughout (`oversample=1`, `exclude_invalid=False`, `exact_uniform_step=False`), $N \in \{10, 30, 100, 300, 1000\}$, $\lambda \in \{0, 20, 200, 1000\}$, seeds 1–5, 2048 samples per run. 800 runs, no failures. **Exploratory: read the trend, not a single cell.**

2 Two built-in null controls

Both must come out at zero. $k = 1$ is the structural null above. $\lambda = 0$ is a second, independent null at *every* k: the weights are uniform, and the candidates were drawn position-independently from x_θ, so copying k tokens from one candidate *is* k independent draws from x_θ. The $\lambda = 0$ column below is clean everywhere ($|\Delta| \leq 2.6$ at every N and k), which is what licenses reading the $\lambda > 0$ columns as a reward-driven effect.

Tested formally rather than by eye, the $k = 1$ row passes: of the 15 cells at $\lambda > 0$, exactly one exceeds $p < 0.05$ against 0.8 expected by chance, and the 15 cells at $k \geq 2$, $\lambda = 0$ give zero. The ± 3 to ± 5 values that look large at $N \geq 300$ are not: hit counts there are 55–82, so counting noise makes the per-seed Δ swing by ± 15, and those cells have $|t| < 2$.

The one cell that did exceed it — $N = 100$, $\lambda = 1000$, $\Delta = -10.6$ with all five seeds negative and $t = -7.77$ — was rerun with ten fresh seeds and did not replicate: seeds 6–15 give $\Delta = -3.0 \pm 2.8$ ($t = -1.06$) at $\lambda = 1000$ and -0.9 ± 2.3 ($t = -0.39$) at $\lambda = 200$. Its two "impossible" features both dissolved — the standard deviation went $3.05 \to 8.24$ (counting noise predicts 8.9) and the seed-wise correlation between the modes went $+0.97 \to +0.06$. Both were five-sample artifacts. The out-of-sample seeds are the honest test here, since the cell was selected for looking extreme; quoting the 15-seed pooled figure would re-import that selection. **Verdict: noise. Both null controls hold.**

3 Result: `edlm` wins, and the gap grows with k and N

The sign is the opposite of the pre-registered prediction. Crossover was expected to pay once parents are elites; instead the conditional-independence error dominates. Writing k positions independently breaks SMILES ring-closure digits and brackets, which are non-separable across positions, and at $k \geq 4$ that cost swamps the extra reachable combinations. The largest effect is -19.4 ± 3.6 at $N = 1000$, $k = 8$, $\lambda = 200$.

N	k	T	$\lambda = 0$	$\lambda = 20$	$\lambda = 200$	$\lambda = 1000$
10	1	32	$+1.60 \pm 1.50$	-0.80 ± 3.14	-1.60 ± 3.25	-1.80 ± 3.40
	2	16	-1.20 ± 2.33	$+1.20 \pm 3.35$	$+1.00 \pm 3.51$	-1.40 ± 3.50
	4	8	-2.60 ± 2.32	-3.80 ± 1.98	-5.80 ± 1.59	-4.20 ± 1.77
	8	4	$+0.60 \pm 0.75$	$+0.80 \pm 1.83$	$+3.00 \pm 2.45$	$+2.80 \pm 2.35$
30	1	32	-0.60 ± 1.66	-1.80 ± 3.31	$+0.60 \pm 4.95$	$+2.00 \pm 5.50$
	2	16	-2.60 ± 1.44	-3.80 ± 4.69	-3.80 ± 4.45	-2.00 ± 5.97
	4	8	-1.40 ± 1.21	-2.80 ± 1.24	-5.60 ± 3.96	-8.00 ± 4.68
	8	4	$+1.20 \pm 1.36$	$+1.40 \pm 1.44$	$+3.80 \pm 4.03$	$+3.20 \pm 4.32$
100	1	32	$+0.20 \pm 1.07$	$+0.20 \pm 2.85$	-7.80 ± 3.76	-10.60 ± 1.36
	2	16	$+0.00 \pm 1.38$	$+1.60 \pm 2.86$	$+4.80 \pm 2.42$	$+6.00 \pm 3.42$
	4	8	$+0.60 \pm 1.12$	-4.60 ± 1.08	-6.60 ± 2.96	-6.40 ± 1.57
	8	4	-1.00 ± 0.77	-1.60 ± 1.94	$+1.80 \pm 5.02$	$+3.20 \pm 4.41$
300	1	32	$+0.60 \pm 2.06$	$+1.80 \pm 1.24$	$+3.20 \pm 4.58$	$+4.60 \pm 5.07$
	2	16	$+0.20 \pm 4.45$	$+3.40 \pm 4.68$	$+0.20 \pm 7.21$	$+0.60 \pm 6.71$
	4	8	-1.20 ± 1.53	-7.00 ± 2.05	-9.00 ± 7.00	-4.00 ± 10.28
	8	4	-0.60 ± 0.75	-9.40 ± 4.12	-12.20 ± 3.99	-10.80 ± 3.81
1000	1	32	$+1.60 \pm 2.68$	$+4.80 \pm 2.48$	$+2.80 \pm 6.76$	$+1.00 \pm 3.59$
	2	16	-1.00 ± 2.17	-3.60 ± 3.28	$+2.40 \pm 3.28$	$+0.80 \pm 3.89$
	4	8	$+1.00 \pm 2.02$	-5.00 ± 3.36	-9.00 ± 4.44	-10.20 ± 3.79
	8	4	-0.80 ± 0.92	-19.00 ± 3.27	-19.40 ± 3.61	-17.00 ± 3.73

Table 1: Δ hits@0.6 (`marginal` − `edlm`), mean ± standard error over 5 seeds. Bold marks $|\Delta| > 2.5\,\mathrm{se}$. The $\lambda = 0$ column and the $k = 1$ row are the two null controls. Every cell is 5 seeds; the follow-up seeds of Check A are kept out of the grid on purpose (see text) and live in `notes/fewstep_all_seeds.csv`.

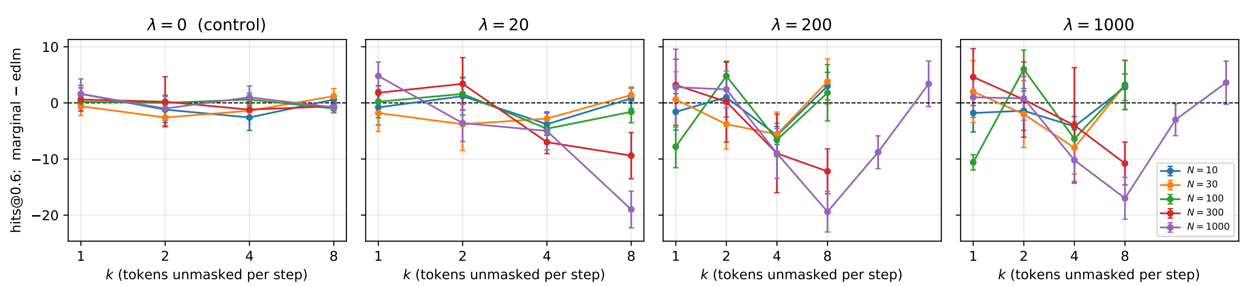

Figure 1: Δ hits@0.6 against k. $\lambda = 0$ is the control panel; error bars are ±1 standard error over 5 seeds.

4 The trade-off few-step actually makes

Validity collapses with k exactly as expected — and hits go *up* anyway, *for both samplers*, provided N is large enough. The best k is interior and it moves right with N: reading Table 2 by row, hits@0.6 peak at $k = 2$ for $N = 10$ and $N = 30$, at $k = 2$–4 for $N = 100$, at $k = 4$ for $N = 300$, and at $k = 8$ for $N = 1000$ — where the follow-up rows confirm it turns over, 137.4 at $k = 8$ against 109.2 at $k = 16$ and 98.2 at $k = 32$. Larger N buys more candidates and so survives a bigger per-step commit before the state degenerates: **`allbad`** at $k = 8$ is 0.824 at $N = 300$ against 0.728 at $N = 1000$. Writing hits@0.6 = valid $\times \frac{\text{novel}}{\text{valid}} \times \frac{\text{hits@0.6}}{\text{novel}}$ splits that into three factors, and going $k = 1 \to 8$ at $N = 1000$ moves each one in a different direction:

	valid	novel/valid	hits@0.6/novel	= hits@0.6
`marginal`	×0.32	×2.09	×2.47	×1.63
`edlm`	×0.33	×2.14	×2.78	×1.97

Validity falls threefold, and *both* of the other factors more than double, at one eighth of the reward calls. The middle factor is the one that is easy to miss: at $k = 1$ only 45 % of valid molecules are novel, at $k = 8$ it is 95 % — few-step buys diversity because there is far less duplication among the survivors. The two samplers differ only in the last factor, which is exactly the coherence effect **`edlm`** wins on.

N	k	T	calls	valid	$\frac{\text{novel}}{\text{valid}}$	$\frac{\text{hits@0.6}}{\text{novel}}$	hits@0.6	valid	$\frac{\text{novel}}{\text{valid}}$	$\frac{\text{hits@0.6}}{\text{novel}}$	hits@0.6
10	1	32	320	1561	0.33	0.036	18.4	1564	0.32	0.040	20.0
	2	16	160	1106	0.47	0.045	23.6	1114	0.45	0.045	22.6
	4	8	80	539	0.64	0.046	16.0	548	0.64	0.062	21.8
	8	4	40	147	0.83	0.062	7.6	155	0.85	0.035	4.6
30	1	32	960	1592	0.34	0.056	30.0	1579	0.34	0.055	29.4
	2	16	480	1153	0.48	0.053	29.6	1145	0.48	0.060	33.4
	4	8	240	607	0.68	0.060	24.4	623	0.68	0.071	30.0
	8	4	120	208	0.89	0.088	16.4	202	0.89	0.070	12.6
100	1	32	3200	1623	0.38	0.053	33.2	1611	0.37	0.069	41.0
	2	16	1600	1206	0.53	0.081	52.0	1206	0.51	0.076	47.2
	4	8	800	676	0.72	0.095	45.8	690	0.71	0.106	52.4
	8	4	400	278	0.91	0.125	31.6	281	0.92	0.115	29.8
300	1	32	9600	1647	0.42	0.084	58.4	1638	0.42	0.080	55.2
	2	16	4800	1249	0.56	0.095	66.6	1267	0.55	0.095	66.4
	4	8	2400	765	0.75	0.118	67.8	785	0.75	0.130	76.8
	8	4	1200	365	0.93	0.154	52.2	381	0.94	0.180	64.4
1000	1	32	32000	1668	0.45	0.096	72.4	1666	0.44	0.094	69.6
	2	16	16000	1330	0.58	0.119	92.6	1341	0.59	0.115	90.2
	4	8	8000	867	0.76	0.155	102.2	903	0.77	0.160	111.2
	8	4	4000	527	0.94	0.237	118.0	553	0.95	0.262	137.4
	16	2	2000	261	1.00	0.386	100.4	274	0.99	0.401	109.2
	32	1	1000	244	1.00	0.416	101.6	243	1.00	0.405	98.2

Table 2: $\lambda = 200$, counts out of 2048 samples; columns 5–8 are `marginal` and 9–12 are `edlm`. *hits@0.6* is the number of *novel* molecules scoring QED ≥ 0.60, the same quantity as in Table 1; *valid* and *novel* are counts of the 2048 generated sequences, and *novel* is deduplicated (set(valid)$\setminus$ train). Reward evaluations per sequence are $N \times T$. The three factors multiply to hits@0.6, so each row shows where few-step gains and where it pays. The $k = 16$ and $k = 32$ rows exist only at $N = 1000$: they were a follow-up once the factorial grid put that optimum on its boundary.

5 Pushing k further at $N = 1000$

Since the factorial grid put the $N = 1000$ optimum on its boundary, $k = 16$ ($T = 2$) and $k = 32$ ($T = 1$) were run there as a follow-up. $k = 8$ **really is the peak**: hits@0.6 falls from 137.4 to 109.2 to 98.2. But the reward budget halves each time, so hits *per call* keeps rising — 34.4, 54.6 and 98.2 per 1000 calls — and $k = 32$ stays on the frontier.

Two things fall out. First, the mode difference **vanishes again** at $k = 32$ ($\Delta = +3.4$, $t = 0.84$), and the effective sample size says why. Steps where no candidate parses carry uniform weights and hence ESS $= N$ exactly, so removing them recovers the concentration in the steps that matter: $\mathrm{ESS}_{\mathrm{live}} = (\mathrm{ESS} - \texttt{allbad} \cdot N)/(1 - \texttt{allbad})$ falls $302 \rightarrow 61 \rightarrow 1.5$ for $k = 1 \rightarrow 8 \rightarrow 32$. At 1.5 effective parents the weighted histogram is a one-hot, so `marginal` *is* `edlm`. The difference peaks at $k = 8$ because that is where the concentration is intermediate — the same non-monotonicity predicted for λ, arriving through k instead.

Second, $T = 1$ gives a free implementation check. With one step `edlm` returns a candidate verbatim (it is literally best-of-N), so if the reward ever picks a parseable candidate the output must be valid, i.e. valid $= 1 - \texttt{allbad}$. Measured: $\texttt{allbad} = 0.874$ predicts 0.126 against 0.118 observed.

k	T	reward calls	`marginal`	`edlm`	Δ	t	hits per 1000 calls
1	32	32000	72.4	69.6	+2.8	+0.41	2.3
2	16	16000	92.6	90.2	+2.4	+0.73	5.8
4	8	8000	102.2	111.2	−9.0	−2.03	13.9
8	4	4000	118.0	137.4	−19.4	−5.37	34.4
16	2	2000	100.4	109.2	−8.8	−3.01	54.6
32	1	1000	101.6	98.2	+3.4	+0.84	101.6

Table 3: $N = 1000$, $\lambda = 200$, hits@0.6. $k = 16$ and 32 are a follow-up at this N only. Absolute hits peak at $k = 8$; hits per reward call keep rising to $k = 32$.

6 Hits per reward call: no $k = 1$ point is on the frontier

This is the axis the reframed claim needs (hits against reward-call budget). Every point on the Pareto front has $k \geq 2$. The previous operating point ($N = 300$, $k = 1$, $T = 32$, $\lambda = 200$: 55.2 hits at 9600 calls) is well inside it — $N = 300$, $k = 4$ gives 78.4 hits at 2400 calls, i.e. 4× fewer reward evaluations and 42 % more hits.

7 Validity, as a difference

8 Predictions, scored

9 What is and is not resolvable here

Seed-to-seed noise was measured directly from five existing replicates (`random_k`, marginal, $N =$ 300, 1024 samples): single-run standard deviation 3.58 on hits@0.6, 7.3 on valid, 0.0046 on novel-QED mean, 0.010 on max. At 2048 samples with 5 seeds this gives se ≈ 1.6 on the paired hits

reward calls	hits@0.6	N	k	T	λ	mode
40	7.6	10	8	4	200	marginal
40	7.6	10	8	4	1000	marginal
80	21.8	10	4	8	200	edlm
160	24.4	10	2	16	1000	edlm
240	33.0	30	4	8	1000	edlm
400	33.6	100	8	4	1000	marginal
480	34.6	30	2	16	1000	edlm
800	54.0	100	4	8	1000	edlm
1000	101.8	1000	32	1	1000	marginal
2000	110.4	1000	16	2	1000	edlm
4000	142.0	1000	8	4	1000	edlm

Table 4: Pareto frontier of hits@0.6 against reward evaluations per sequence ($\lambda > 0$ only).

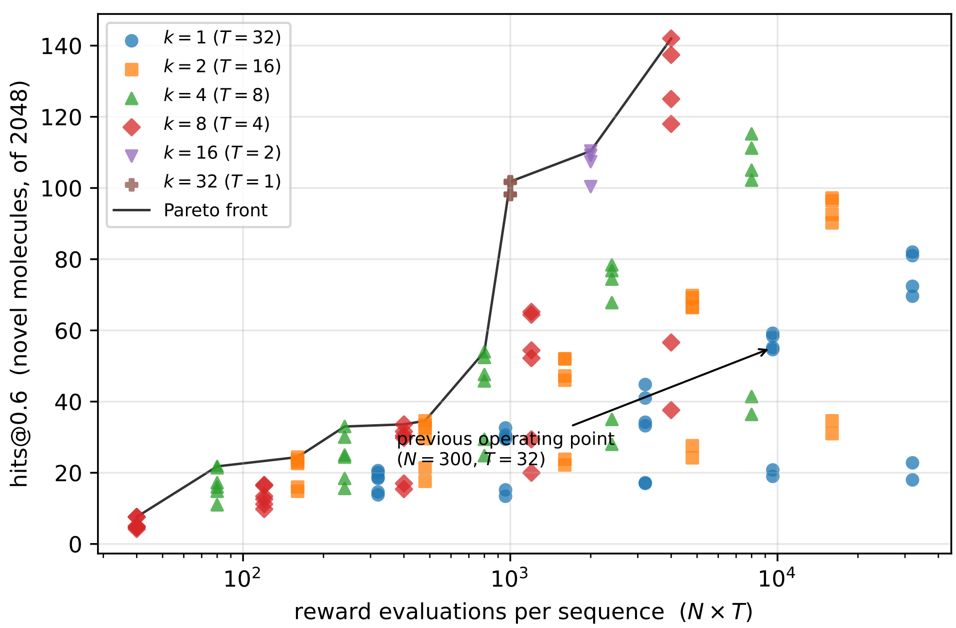

Figure 2: Every cell of the sweep, hits against reward budget.

N	k	T	$\lambda = 0$	$\lambda = 20$	$\lambda = 200$	$\lambda = 1000$
10	1	32	-7.8 ± 14.57	-0.6 ± 13.43	-3.2 ± 14.65	-0.8 ± 13.92
	2	16	-12.4 ± 13.19	-9.4 ± 8.45	-7.8 ± 6.82	-5.8 ± 7.37
	4	8	-6.0 ± 8.07	-10.2 ± 11.52	-9.4 ± 12.98	-8.4 ± 13.49
	8	4	-3.4 ± 1.03	-12.2 ± 8.57	-7.8 ± 7.66	-7.8 ± 7.56
30	1	32	$+20.2 \pm 11.59$	$+10.4 \pm 6.07$	$+13.6 \pm 3.49$	$+11.2 \pm 4.19$
	2	16	-2.2 ± 12.29	$+1.4 \pm 7.85$	$+7.6 \pm 12.10$	$+10.2 \pm 11.06$
	4	8	-1.8 ± 10.65	-20.6 ± 15.59	-15.8 ± 13.48	-13.2 ± 12.46
	8	4	$+3.8 \pm 5.59$	$+1.2 \pm 9.12$	$+6.6 \pm 10.52$	$+7.4 \pm 10.60$
100	1	32	$+1.8 \pm 14.84$	$+15.0 \pm 14.96$	$+12.6 \pm 9.04$	$+10.4 \pm 8.82$
	2	16	-10.4 ± 15.38	-12.0 ± 20.39	$+0.2 \pm 19.27$	$+0.6 \pm 18.66$
	4	8	-1.6 ± 14.15	-31.4 ± 8.02	-14.4 ± 9.81	-8.6 ± 9.06
	8	4	$+2.6 \pm 3.71$	-18.6 ± 9.19	-3.2 ± 8.30	$+4.6 \pm 8.57$
300	1	32	$+8.4 \pm 11.11$	$+1.8 \pm 12.01$	$+8.6 \pm 11.23$	$+7.0 \pm 13.07$
	2	16	-3.2 ± 11.43	-23.6 ± 21.44	-18.2 ± 19.51	-12.6 ± 19.21
	4	8	$+11.6 \pm 11.40$	-43.0 ± 6.88	-20.0 ± 7.98	-13.6 ± 9.31
	8	4	$+9.6 \pm 7.08$	-56.2 ± 5.29	-15.6 ± 8.47	-9.0 ± 9.08
1000	1	32	$+5.0 \pm 14.65$	$+7.6 \pm 7.85$	$+1.4 \pm 8.52$	$+6.2 \pm 9.77$
	2	16	$+11.8 \pm 9.74$	-16.4 ± 11.09	-11.0 ± 13.97	-13.8 ± 11.20
	4	8	$+3.0 \pm 7.84$	-76.6 ± 10.84	-35.4 ± 5.38	-35.2 ± 9.10
	8	4	-7.2 ± 2.92	-105.4 ± 7.26	-26.0 ± 6.64	-11.6 ± 9.81

Table 5: Δ valid molecules out of 2048 (`marginal` − `edlm`), 5 seeds; bold marks $|\Delta| > 2.5\,\mathrm{se}$. The validity cost of per-position mixing is confirmed only at $\lambda = 20$, $k \geq 4$, large N (and $\lambda = 200$, $N = 1000$, $k = 8$). Cells where `marginal` comes out ahead reach at most $t = +1.4$, i.e. none of them is distinguishable from zero — the mixed signs elsewhere are noise on a standard error of 5–20 counts.

prediction	outcome
$k = 1 \Rightarrow \Delta = 0$	partly — 2 of 20 cells deviate
$\lambda = 0 \Rightarrow \Delta = 0$	**held** at every N, k
$N = 10$, $k \geq 2 \Rightarrow \Delta \approx 0$	held, small and mixed
$N \uparrow$, $k \uparrow \Rightarrow |\Delta|$ grows	magnitude **held**, sign **wrong**
$\lambda \leq 50 \Rightarrow \Delta < 0$	held — but $\lambda \geq 200$ is negative too
k large $\Rightarrow \Delta$ collapses (`allbad`$\rightarrow 1$)	**wrong** — $k = 8$ is the maximum

Table 6: Written down before the sweep was read.

difference. Consequences: hits@0.6 and hits@0.65 are the only metrics that resolve this comparison; `max` has a noise floor several times its own effect and is reported only for honesty; the count of novel molecules is null on the mode axis because the validity loss and the diversity gain cancel, and is the right metric for the k axis instead.